



Capital University

Faculty of Engineering
Computer & Systems Engineering Department

University Management System

Database Design Documentation

Entity Relationship Diagram & Schema Mapping

Team Members

Abdelrahman Aboud
Mohamed Amgad
Mohamed Emad
Ahmed Hussein
Mazen Ahmed

Under the Supervision of
Eng. Nancy

1. Introduction

This document provides a comprehensive description of the database schema for the University Management System (UMS). It covers all entities, their attributes, primary and foreign keys, and the relationships between them as derived from the Entity Relationship Diagram (ERD) and schema mapping.

The system is designed to manage the core academic operations of a university, including user management (students, doctors, and administrators), course and class administration, lecture scheduling, learning materials, grade tracking, attendance monitoring, and an integrated Q&A functionality for academic interaction.

The database was designed following standard relational database principles, with clear separation of concerns across entities, normalized data structures, and well-defined foreign key relationships to ensure data integrity and consistency across the system.

Document Objectives

- Describe the complete database schema including all tables, attributes, and data types.
- Define all primary keys, foreign keys, and entity relationships.
- Document ISA (specialization) relationships between the User base entity and its subtypes.
- Serve as a reference guide for developers, database administrators, and project stakeholders.

2. System Overview

The University Management System is a relational database-backed platform built to digitize and streamline all key academic processes within a university environment. It supports role-based access for three user types — **Students**, **Doctors**, and **Administrators** — all inheriting from a shared **User** entity via ISA specialization. The system is structured around **14 entities** organized across five functional categories:

- **Core User Entities:** User, Student, Doctor, Admin — manage authentication and role-specific profile data.
- **Academic Structure:** Department, Courses, Class, Lecture — organize academic offerings, scheduling, and delivery.
- **Content & Assessment:** Material, Grades, Attendance — handle learning resources, NFC-based attendance, and grade records.
- **Interaction:** Questions, Answer — enable students to post academic questions and receive answers from peers or doctors.
- **Relationship Tables:** Enrollment, Work In, Offers — resolve many-to-many relationships between core entities.

3. Entity Descriptions

3.1 User

The base entity representing all system users. Student, Doctor, and Admin entities extend this entity through specialization (ISA relationship).

Attribute	Data Type	Key	Description
<u>User_ID</u>	INT	PK	Unique identifier for each user
Password	VARCHAR(255)	—	Hashed password for authentication
F_Name	VARCHAR(50)	—	User's first name
L_Name	VARCHAR(50)	—	User's last name
Email	VARCHAR(100)	UNIQUE	University email address
Account_Status	ENUM	—	Active / Inactive / Suspended
Role	ENUM	—	Student / Doctor / Admin
Document	VARCHAR(255)	—	Path to identity document file
Image_Url	VARCHAR(255)	—	URL/path to user profile picture

3.2 Student

Extends User. Represents enrolled students with academic and payment attributes.

Attribute	Data Type	Key	Description
<u>User_ID (FK)</u>	INT	PK / FK → User	Inherits User_ID as primary key
Academic_Level	INT	—	Year/Level in academic program (e.g., 1–4)
Payment_Status	ENUM	—	Paid / Unpaid / Partial
NFC_Tag_ID	VARCHAR(50)	UNIQUE	NFC tag for physical attendance check-in
SSN	VARCHAR(20)	UNIQUE	National ID / Student Social Security Number
Dept_ID (FK)	INT	FK → Department	Department the student belongs to
Total_Hours	INT	—	Total hours available for student
Total_GPA	DECIMAL(4,2)	—	Cumulative GPA of the student

3.3 Doctor

Extends User. Represents academic staff / instructors.

Attribute	Data Type	Key	Description
<u>User_ID</u> (FK)	INT	PK / FK → User	Inherits User_ID as primary key
Specialization	VARCHAR(100)	—	Doctor's area of academic specialization

3.4 Admin

Extends User. Represents system administrators with configurable permission levels.

Attribute	Data Type	Key	Description
<u>User_ID</u> (FK)	INT	PK / FK → User	Inherits User_ID as primary key
Permissions_Level	INT / ENUM	—	Level of system access (e.g., 1=read, 2=write, 3=super)

3.5 Department

Represents an academic department within the university. A department can offer multiple courses and employ multiple doctors.

Attribute	Data Type	Key	Description
<u>Dept_ID</u>	INT	PK	Unique department identifier
Dept_Name	VARCHAR(100)	UNIQUE	Name of the department
Doctor_ID (FK)	INT	FK → Doctor	Department head or associated doctor
Permission	VARCHAR(100)	—	Department-level access/permission configuration

3.6 Courses

Represents course definitions. A course is a general blueprint offered by departments.

Attribute	Data Type	Key	Description
<u>Course Code</u>	VARCHAR(20)	PK	Unique code identifying the course (e.g., CS101)
Name	VARCHAR(100)	—	Full name of the course
Credit_Hours	INT	—	The total credits per course

3.7 Class

A specific instance of a course offered in a particular semester and level, assigned to a doctor.

Attribute	Data Type	Key	Description
<u>Class ID</u>	INT	PK	Unique class section identifier
Level	INT	—	Academic level this class targets
Semester	VARCHAR(20)	—	Semester offered (e.g., Fall 2025)
Course_Code (FK)	VARCHAR(20)	FK → Courses	The course this class belongs to
Doctor_ID (FK)	INT	FK → Doctor	Doctor responsible for this class
Capacity	INT	—	The max number of students per class

3.8 Lecture

Represents a scheduled lecture session within a class. Can be physical (room-based) or online (meeting link).

Attribute	Data Type	Key	Description
<u>Lec ID</u>	INT	PK	Unique lecture identifier
Day	VARCHAR(15)	—	Day of the week (e.g., Monday)
Time	TIME	—	Scheduled start time of the lecture
Room_ID	VARCHAR(20)	—	Physical room or lab identifier
Meeting_Link	VARCHAR(255)	—	Online meeting URL (if virtual)
Type	ENUM	—	Lecture / Section / Lab / Online
Class_ID (FK)	INT	FK → Class	The class this lecture belongs to

3.9 Material

Stores learning materials (files, links, or AI-generated summaries) associated with a specific lecture.

Attribute	Data Type	Key	Description
<u>Material_ID</u>	INT	PK	Unique material record identifier
<u>Lec_ID</u> (FK)	INT	FK → Lecture	Lecture this material belongs to
Name	VARCHAR(100)	—	Display name of the material
URL	VARCHAR(255)	—	Link to external resource or video
Document	VARCHAR(255)	—	File path to uploaded document (PDF, PPT, etc.)
Summarize	TEXT	—	AI-generated or manual summary of the material

3.10 Grades

Stores assessment grades for a student in a specific class, broken down by component.

Attribute	Data Type	Key	Description
<u>Grade_ID</u>	INT	PK	Unique grade record identifier
Type	ENUM	—	Grade category type
Generate_At	DATETIME	—	Timestamp when grade was recorded
Attendance	DECIMAL(5,2)	—	Attendance score/percentage
Practical	DECIMAL(5,2)	—	Practical/lab grade
Project	DECIMAL(5,2)	—	Project grade
Midterm	DECIMAL(5,2)	—	Midterm exam grade
Final	DECIMAL(5,2)	—	Final exam grade
<u>User_ID</u> (FK)	INT	FK → Student	Student this grade belongs to
<u>Class_ID</u> (FK)	INT	FK → Class	Class the grade is for
<u>Doctor_ID</u> (FK)	INT	FK → Doctor	Doctor who entered the grade
GPA	DECIMAL(4,2)	—	Overall Performance Average (OPA) / GPA for the class

3.11 Attendance

Records individual student attendance events per lecture, supporting NFC-based verification.

Attribute	Data Type	Key	Description
<u>Attendance_ID</u>	INT	PK	Unique attendance event identifier
<u>User_ID</u> (FK)	INT	FK → Student	Student who attended
<u>Lec_ID</u> (FK)	INT	FK → Lecture	Lecture the attendance belongs to
Time	DATETIME	—	Exact timestamp of check-in
Is_Verified	BOOLEAN	—	True if verified via NFC; False if manual

3.12 Questions

Allows students to post questions related to a specific class. Questions can be directed to a doctor.

Attribute	Data Type	Key	Description
<u>Questions_ID</u>	INT	PK	Unique question identifier
Text	TEXT	—	The question content
User_ID (FK)	INT	FK → Student	Student who posted the question
Class_ID (FK)	INT	FK → Class	Class context of the question
Doctor_ID (FK)	INT	FK → Doctor	Doctor the question is directed to

3.13 Answer

Stores answers to student questions. Answers can come from doctors or other students.

Attribute	Data Type	Key	Description
<u>Answer_ID</u>	INT	PK	Unique answer identifier
<u>Questions_ID</u> (FK)	INT	FK → Questions	The question being answered
Text	TEXT	—	The answer content
Time	DATETIME	—	Timestamp when answer was posted
Doctor_ID (FK)	INT	FK → Doctor	Doctor who answered (if applicable)
User_ID (FK)	INT	FK → Student	Student who answered (if applicable)

4. Relationship Tables (Junction Tables)

4.1 Enrollment

Many-to-many relationship between Student and Class. Records which students are enrolled in which class sections.

Attribute	Data Type	Key	Description
<u>Class_ID</u> (FK)	INT	PK / FK → Class	The class being enrolled in
<u>User_ID</u> (FK)	INT	PK / FK → Student	The student enrolling

Primary Key: Composite (Class_ID, User_ID)

4.2 Work In

Many-to-many relationship between Doctor and Department. Records which departments a doctor works in.

Attribute	Data Type	Key	Description
<u>Doctor_ID</u> (FK)	INT	PK / FK → Doctor	The doctor working in the department
<u>Dept_ID</u> (FK)	INT	PK / FK → Department	The department

Primary Key: Composite (Doctor_ID, Dept_ID)

4.3 Offers

Many-to-many relationship between Department and Courses. Records which courses are offered by which departments.

Attribute	Data Type	Key	Description
<u>Course_Code</u> (FK)	VARCHAR(20)	PK / FK → Courses	The offered course
<u>Dept_ID</u> (FK)	INT	PK / FK → Department	The department offering it

Primary Key: Composite (Course_Code, Dept_ID)

4.4 Recorded Course

The Names of subjects selected by the user.

Attribute	Data Type	Key	Description
<u>User_ID</u> (FK)	INT	PK / FK → Student	The student recording
<u>Course_Code</u> (FK)	INT	PK / FK → Courses	The Recorded courses for students

Primary Key: Composite (Course_Code, User_ID)

5. Entity Relationships Summary

Relationship	From Entity	To Entity	Type	Via / Notes
User → Student	User	Student	1 : 1	ISA / specialization
User → Doctor	User	Doctor	1 : 1	ISA / specialization
User → Admin	User	Admin	1 : 1	ISA / specialization
Student ↔ Department	Student	Department	N : 1	Dept_ID FK on Student
Doctor ↔ Department	Doctor	Department	M : N	Work In junction table
Department ↔ Courses	Department	Courses	M : N	Offers junction table
Courses → Class	Courses	Class	1 : N	Course_Code FK on Class
Doctor → Class	Doctor	Class	1 : N	Doctor_ID FK on Class
Class → Lecture	Class	Lecture	1 : N	Class_ID FK on Lecture
Lecture → Material	Lecture	Material	1 : N	Lec_ID FK on Material
Student ↔ Class	Student	Class	M : N	Enrollment junction table
Student → Attendance	Student	Attendance	1 : N	User_ID FK on Attendance
Lecture → Attendance	Lecture	Attendance	1 : N	Lec_ID FK on Attendance
Student → Grades	Student	Grades	1 : N	User_ID FK on Grades
Student → Questions	Student	Questions	1 : N	User_ID FK on Questions
Questions → Answer	Questions	Answer	1 : N	Questions_ID FK on Answer
Doctor → Answer	Doctor	Answer	1 : N	Doctor_ID FK on Answer
Student → Answer	Student	Answer	1 : N	User_ID FK on Answer
Student ↔ Recorded	Student	Courses	M : N	Recorded junction table (User_ID, Subject)

6. Key Notation Legend

Symbol / Color	Meaning
PK (Blue cell)	Primary Key – uniquely identifies a record in the table
FK (Dark red text)	Foreign Key – references a primary key in another table
UNIQUE	Attribute must be unique across all records
M : N	Many-to-Many relationship, resolved via a junction table
1 : N	One-to-Many relationship
ISA	Specialization / Inheritance — child entity extends the parent User entity